SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1007	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 1 – **Scenario-Based Requirement Analysis**
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	28/07/2026	**Duration:**	60 minutes

Code of Conduct

*I **TARUNIKA S(192511024**) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: TARUNIKA S

ANNEXURE A - Scenario-Based Requirement Analysis

Smart High-Speed Rail Passenger Experience Platform

1. Introduction

The Smart High-Speed Rail Passenger Experience Platform is a digital system that enables online ticket booking, real-time train tracking, onboard services, safety alerts, and journey management through web and mobile applications. It improves passenger experience while enhancing railway operational efficiency.

2. Problem Description

Current high-speed rail systems use separate ticketing and passenger management systems, causing booking delays, limited real-time updates, manual boarding processes, and poor access to onboard services. This reduces passenger satisfaction and operational efficiency.

3. Scope of the System

The Smart High-Speed Rail Passenger Experience Platform enables passengers to book tickets, access QR-based ticket verification, track trains in real time, and receive journey updates. It also offers onboard services such as food ordering and entertainment while providing staff with passenger information and administrators with analytics through integrated railway systems.

4. Actors

- **Passenger** – Books tickets, tracks trains, uses onboard services, and gives feedback.
- **Train Crew/Station Staff** – Manages boarding, seating, and passenger assistance.
- **System Administrator** – Monitors the platform and manages reports.
- **Payment Gateway** – Processes ticket payments and refunds.
- **Railway Operations System** – Provides live train status, schedules, and delay updates.

5. Functional Requirements

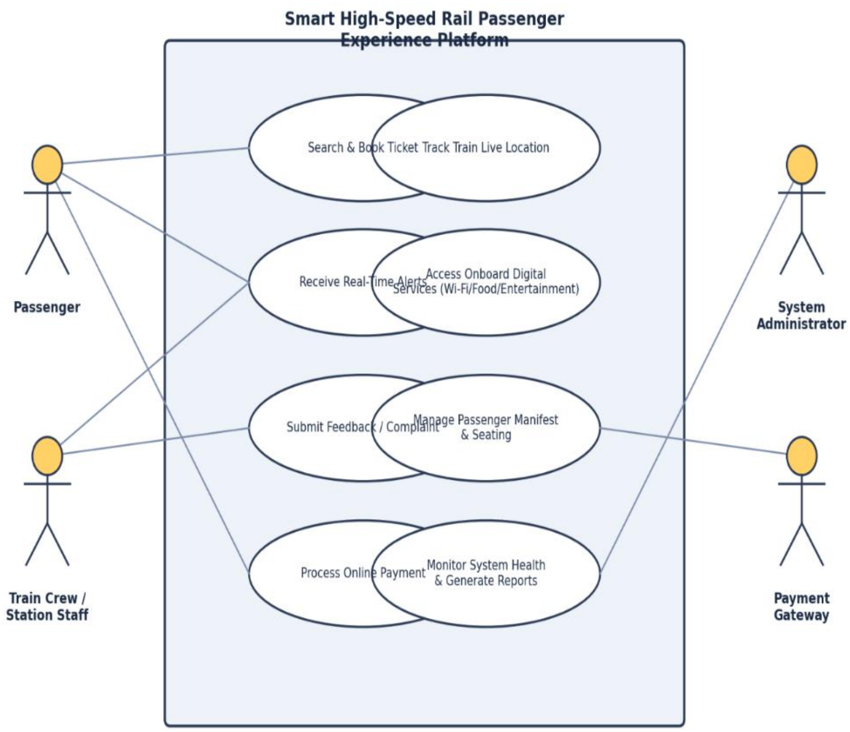
- FR1: Search, book, and cancel train tickets.
- FR2: Select seats and generate QR tickets.
- FR3: View live train location and status.
- FR4: Receive delay and platform change alerts.
- FR5: Access Wi-Fi, food, and entertainment services.
- FR6: Submit journey feedback and complaints.
- FR7: View passenger list and seat occupancy.
- FR8: Report onboard incidents and emergencies.
- FR9: Process secure payments and refunds.
- FR10: Generate analytics and performance reports.

6. Non-Functional Requirements

- **Performance & Scalability:** Fast train search, live updates, and support for high user traffic.
- **Security & Reliability:** Secure data encryption, role-based access, and 99.9% system uptime.
- **Usability & Availability:** User-friendly, multilingual interface with 24/7 service availability.
- **Maintainability:** Modular system design for easy updates and future enhancements.

7. Use Case Diagram

The diagram below summarizes the primary actors and their key interactions with the platform.



9. Assumptions & Constraints

Assumptions

- Users have internet access via mobile/web.
- Railway system provides real-time API data.
- Wi-Fi/connectivity infrastructure is available.

Constraints

- Integrate with existing railway systems.
- Ensure 24/7 availability with backup support.
- Limited budget for pilot rollout first.